

Towards an Active Inference Personality Framework: The Core Prior Cascade Model of the Self

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The Core Prior Cascade (CPC) Model is a hierarchical generative account of personality and individual differences grounded in the Free Energy Principle and active inference. In contrast to descriptive psychometric frameworks that identify behavioural phenotypes, the CPC Model proposes that observable personality traits represent the downstream expressions of deeply entrenched probabilistic priors: hierarchically organised beliefs governing an agent's expectations about environmental volatility, action tractability, and self-world boundary structure. Three tiers of prior structure are identified: Existential Hyperpriors, encoding global parameters of volatility, agency, and temporal self-extension; Boundary and Relational Priors, governing Markov blanket permeability and sensory grounding; and Epistemic Meta-Priors, regulating error-resolution strategies and self-model coherence. From this hierarchical structure, five latent factors are hypothesised. These factors (Epistemic Plasticity, Cybernetic Stability, Blanket Permeability, Temporal Extension, and Self-Model Stability) are argued to constitute the generative mechanisms underlying the phenotypic variance captured by existing descriptive models including the Big Five, behavioural inhibition and activation systems, attachment theory, and sensation seeking. The CPC Model is presented as a principled, mechanistic complement to existing descriptive frameworks, offering testable predictions regarding the hierarchical structure of individual differences and differential psychopathological vulnerability.

Impact Statement

The Core Prior Cascade Model provides a mechanistic, mathematically grounded account of how hierarchical prior structure generates the full range of observable personality phenotypes, offering a principled bridge between the Free Energy Principle and the scientific study of individual differences.

Keywords: active inference, free energy principle, personality, individual differences, predictive processing, cybernetics, hyperpriors

Personality psychology faces a persistent explanatory gap. While descriptive frameworks, such as the Big Five model of personality traits (Goldberg, 1990; McCrae & John, 1992), have proven

remarkably successful at organising individual differences in behaviour, affect, and cognition, their structural parsimony comes at a theoretical cost. These models describe what varies across individu-

als but offer limited purchase on why such variation exists or how it is generated and maintained. Comprehensive mechanistic accounts capable of specifying the processes that produce the characteristic patterns we label as Neuroticism, Openness, or Conscientiousness are relatively scarce.

Several generative frameworks have attempted to bridge this descriptive-process divide. The Cognitive-Affective Personality System (CAPS, [Mischel & Shoda, 1995](#)) conceptualises personality as a network of interacting cognitive and affective units, while Whole Trait Theory ([Fleeson & Jayawickreme, 2015](#)) frames traits as density distributions of momentary states. More recently, Cybernetic Big Five Theory (CB5T, [DeYoung, 2015](#)) has offered a functionalist account of traits as parameters of a cybernetic system optimising for distinct goals. Yet, while these models successfully introduce process dynamics, they largely lack a unified mathematical ontology to explain why certain cognitive-affective parameters reliably coalesce into the stable phenotypic attractors we observe.

The Free Energy Principle (FEP, [Friston, 2010](#)) and its process theory, active inference, offer a principled candidate for such a generative account. Under the FEP, all self-organising biological systems minimise the upper bound on their own surprise, formally defined as variational free energy, by maintaining and continuously updating probabilistic generative models of their sensory environment. This minimisation operates through two

complementary pathways: perceptual inference (revising internal models to fit sensory signals) and active inference (acting on the world to bring sensory signals into conformity with predictions). Applied to the domain of personality, this framework implies that characteristic individual differences do not reflect fixed trait properties of the organism but rather the downstream consequences of deeply entrenched probabilistic priors. These are beliefs the agent's predictive architecture treats as near-inviolable constraints on all subsequent inference and action. Personality, on this account, is not a static label but a characteristic cybernetic strategy.

The present paper proposes the Core Prior Cascade (CPC) Model, a hierarchical framework that organises these foundational priors across three tiers of abstraction and influence. At the apex sit *Existential Hyperpriors*, representing the most fundamental assumptions the agent holds about the volatility of its environment, the tractability of prediction error through action, and the ultimate temporal horizon of the self. These global parameters cascade downward to constrain a second tier of *Boundary and Relational Priors*. This middle tier governs the permeability and sensory grounding of the agent's Markov blanket, the statistical boundary demarcating self from environment ([Friston, 2019](#)). These in turn constrain a third tier of *Epistemic Meta-Priors*, governing the agent's characteristic strategies for minimising free energy across the perception-action cycle.

The CPC Model is explicitly positioned not as a replacement for existing descriptive frameworks but as a generative account of the mechanisms underlying the phenotypes those frameworks identify. The Big Five describe individual differences at the level of behavioural output, and the Cybernetic Big Five Theory begins to explain them at the level of cybernetic function. The CPC Model proposes to account for both at the level of the hierarchical prior structure from which such function emerges. It thereby offers testable structural predictions and a principled basis for understanding individual differences in psychopathological vulnerability.

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The Core Prior Cascade Model

The central architectural claim of the CPC Model is that the cascade is not merely correlational. Higher-tier priors do not just co-vary with lower-tier ones; they constrain the solution space available to them. A Tier 1 configuration of extreme volatility and low tractability forecloses certain Tier 3 epistemic strategies. This hierarchical determination is what differentiates the CPC Model from multivariate trait models in which dimensions are statistically correlated but conceptually orthogonal.

A productive conceptual frame for this cascade architecture is that of a *free energy landscape*. This is a topographic surface on which the agent's achievable prior configurations are mapped against their associated expected free energy. Stable personality configurations correspond to local minima acting as *attractor basins* into which the system settles and from which it resists displacement. The depth of a basin reflects the precision weighting assigned to the prior structure at that tier. Deeply entrenched hyperpriors carve deep attractors that demand substantial perturbation to escape. Mild challenges to the generative model produce only local oscillations within the basin. However, more substantial perturbations, such as developmental transitions, significant relational experiences, or sustained therapeutic intervention, may shift the system to an adjacent attractor. Catastrophic destabilisation may temporarily flatten the landscape entirely, transiently liberating the system from habitual prior constraints.

The *valleys* of this landscape are the priors themselves. Characteristic personality configurations represent the stable dynamical states of an agent's free energy landscape. Psychopathological trajectories can be understood as transitions between basins, as trapping within anomalously deep basins, or as the fragmentation of a landscape too shallow to sustain coherent stable states. This framing also accommodates the bidirectional dynamics of the cascade. While higher-tier priors constrain the solution space for lower-tier configurations, bottom-

up neuroplasticity continuously reshapes the landscape through experience and development, gradually deepening or shallowing the attractor basins themselves.

The landscape also has an overall topology that shapes all local dynamics. Death, defined as systemic cessation or the dissolution of the agent's self-organising boundary, is not merely one attractor basin among many but the global minimum. It is the deepest point toward which any finite system will eventually tend. All other personality configurations are metastable states on a landscape that tilts, over sufficiently long timescales, toward this terminus. The terminal prior, described below, governs how consciously and at what distance the agent orients to this underlying structure.

The following sections specify the dimensions at each tier, moving from the most global and abstract to the most proximal and context-sensitive.

Existential Hyperpriors

Existential hyperpriors occupy the apex of the cascade hierarchy. They encode the agent's most fundamental assumptions about the nature of its environment and the ultimate tractability of its predictive project. Because they operate at the highest level of abstraction, perturbations at this tier propagate throughout the entire system. They are described here as "existential" because they concern not merely strategic preferences for error minimisation but the agent's basic orientation to its own existence in time, addressing questions of agency, safety, and finitude.

Volatility Prior

The volatility prior encodes the agent's baseline expectation regarding the rate of change in the environment's hidden states, which is the underlying generative structure the agent's model attempts to track. An agent with a *high volatility prior* expects the environment's generative rules to shift rapidly. This forces a systematic discounting of historical learning and a near-continuous anticipation of pre-

diction error. Past model updates are rapidly deprecated because the agent expects the underlying structure they tracked has already changed. Conversely, an agent with a *low volatility prior* expects the environment to be highly structurally stable over time, affording the accumulation of high-precision long-term models and a characteristic orientation toward order and environmental mastery. The volatility prior constitutes one of the primary Tier 1 parameters because it sets the time horizon over which all downstream inference is meaningful: an agent cannot maintain stable Tier 2 and Tier 3 strategies in an environment it expects to be fundamentally unpredictable.

Tractability Prior

The tractability prior encodes the agent's baseline expectation of whether prediction errors, once encountered, can be resolved through the agent's own active inference. A *high tractability prior* reflects the belief that action reliably reduces error, meaning the gap between the agent's internal model and the sensory world is bridgeable through purposive agency. This prior disposition produces approach behaviour, behavioural activation, and sustained engagement with uncertainty, because the agent expects that exploration will ultimately yield resolvable error.

A *low tractability prior* reflects the expectation that action generates unresolvable error, or that the system simply lacks the capacity to shift external states into alignment with its internal model. This configuration shares close parallels with learned helplessness (Seligman, 1972). Low tractability drives epistemic shielding (the suppression of precise sensory signals to avoid errors that cannot be remedied) and behavioural inhibition as the energetically efficient response to expected futility. The tractability prior is explicitly distinguished from the volatility prior, though the two may correlate. An agent in a genuinely volatile environment may rationally develop low tractability expectations, while an agent with a constitutional low tractability prior may perceive even stable environ-

ments as intractably uncertain.

Irreducible Uncertainty Tolerance

Irreducible uncertainty tolerance is a reactive parameter governing the agent's response when epistemic foraging has failed. This occurs when active sampling of informative sensory states has not resolved a prediction error. It is formally distinct from the tractability prior: where tractability concerns the agent's prior belief about whether action can resolve error, uncertainty tolerance concerns the agent's capacity to sustain functioning when a prediction error proves genuinely unresolvable post hoc.

These parameters can dissociate in important ways. An agent with a high tractability prior may be surprisingly brittle upon encountering genuine irreducibility, having rarely needed to develop tolerance for unresolvable error. Alternatively, an agent with chronically low tractability may develop high baseline tolerance, having habituated to the persistent presence of residual uncertainty. *High uncertainty tolerance* reflects the capacity to accept irreducible error, representing an epistemic humility that permits adequate functioning without forcing premature model updates or false closure. *Low uncertainty tolerance* drives premature cognitive closure, distress in the face of ambiguity, and a tendency to force lower-level models to accommodate incomplete or contradictory data at the cost of model accuracy.

Affective Valence

Affective valence is defined as the characteristic phenomenological tone of the agent's expected free energy signal. Within the FEP, persistent negative affect is interpretable as the felt correlate of chronically elevated expected free energy. The agent continuously anticipates unresolvable prediction errors, and this anticipation is experienced as anxiety, dysphoria, or anhedonia. Affective valence is treated here as an explicit Tier 1 dimension, not merely a correlate of the preceding parameters, because it carries independent predictive

value and constitutes the primary experiential interface through which higher-tier prior structure is registered.

Positive valence reflects low baseline expected free energy, characterized by an affective tone of approach, safety, and reward anticipation associated with a belief that the world is broadly navigable. *Negative valence* reflects chronically elevated expected free energy, leading to persistent threat anticipation, vigilance, or hedonic blunting. This dimension maps directly onto the construct of Neuroticism in descriptive models (Eysenck, 1967; McCrae & John, 1992) and provides the felt experiential counterpart to the tractability prior. On this account, Neuroticism is not a primary trait but the phenomenological consequence of Tier 1 hyperprior configuration, reflecting the experienced signature of low tractability combined with high volatility expectations.

This formulation converges with Barrett’s (2017) Theory of Constructed Emotion, which frames affect as a low-dimensional readout of the body’s allostatic budget, functioning as a continuous barometer of the anticipated cost of maintaining homeostatic and predictive integrity. The CPC Model situates this readout at the apex of the prior hierarchy. Affective valence acts as the phenomenological signature of the agent’s net expected free energy at the level of existential hyperpriors, and is therefore amongst the most conservative and change-resistant dimensions in the framework.

Terminal Prior

The terminal prior governs how the agent models its own ultimate structural dissolution. Within the FEP, mortality salience functions as a hyperprior precision-weighting problem: confrontation with finitude forces the agent to resolve or suppress the prediction error of anticipated self-model cessation (Becker, 1973; Greenberg et al., 1986). The temporal distance over which the agent’s self-model projects itself before anticipating dissolution, here termed the *temporal horizon of the self*, constitutes one of the primary axes of Tier 1 variation.

Three configurations are identified. *Disengagement* reflects an attenuation of the temporal self-model and takes two functionally distinct forms. Passive disengagement involves a diffuse failure of deep temporal planning characterised by generalised low precision on future-directed inference and a fatalistic or anhedonic orientation; the agent drifts rather than actively converging on future states. Active cessation-seeking represents a catastrophically precise prior converging on systemic shutdown as the optimal policy. Here, the expected free energy of all future states is evaluated as unresolvably high, rendering structural cessation the lowest-free-energy attractor. This configuration is the formal active inference analogue of Thanatos, mapping onto clinical suicidal ideation at its extreme.

Integration reflects authentic acceptance of finitude. Expected free energy is balanced across temporal horizons without needing to suppress mortality salience, and the agent optimises for present-horizon goals without either withdrawing from the future or frantically extending the self into it. *Transcendence* involves extending the Markov blanket across time through legacy, whether cultural, symbolic, or biological, such that the agent’s self-model is projected to outlive the physical substrate. Transcendence and Integration may represent two configurations along a continuous dimension of temporal self-extension, while disengagement constitutes a qualitatively distinct attractor state.

Boundary and Relational Priors

Boundary and relational priors govern the expected state of the interface between the agent’s internal model and the external world, establishing the functional properties of the Markov blanket as a dynamic boundary through which inference flows. These priors are constrained by Tier 1: an agent whose hyperprior configuration imposes extreme volatility and low tractability has correspondingly fewer viable configurations available at Tier 2. The three dimensions at this tier concern the gradient of coupling between self and world, the precision allo-

cated to social inference, and the sensory modality that serves as the agent's fundamental ground truth.

Permeability Prior

The permeability prior governs the relative precision weighting the agent assigns to self-referential priors versus exteroceptive and social signals. This is not a question of the Markov blanket becoming literally permeable, as a dissolved blanket takes the subject with it, but of adjusting the gradient of influence across the boundary. It dictates how strongly external states are permitted to update internal ones, and vice versa. This maps onto Bakan's (1966) foundational distinction between agency and communion, and onto the attachment styles described by Bowlby (1969) and Ainsworth: avoidant (high differentiation, low coupling), anxiously-preoccupied (high dissolution, high coupling with low tractability), and secure (calibrated coupling, high tractability).

The *differentiation drive* reflects the expectation that internal states must remain structurally distinct from external states. It is a low-coupling, rigid-blanket configuration associated with drives for autonomy, individuation, and status maintenance. At its extreme, the differentiation drive motivates dominance-seeking as an active inference strategy. Forcing the social environment to synchronise with the agent's internal model achieves error minimisation through environmental subjugation rather than internal revision.

The *dissolution drive* reflects the expectation that internal states should synchronise with external states. This high-coupling configuration is associated with drives for affiliation, merging, and distributing cognitive load across the social group. Agents with a strong dissolution drive are oriented toward cooperative rather than competitive modes of free energy minimisation and are characterised by heightened sensitivity to social prediction errors.

Social Inference Prior

The social inference prior concerns the precision the agent assigns to modelling the hidden states of other agents, representing the computational resources allocated to mentalising. An agent with *high social inference precision* deeply invests in simulating others' internal models, generating rich predictive representations of social others' beliefs, intentions, and affective states. This manifests as heightened empathy and vigilance to social cues, but also as elevated susceptibility to social prediction errors: the more precise the model of another agent, the more costly any violation of that model. An agent with *low social inference precision* treats other agents more as physical environmental variables than as nested cognitive systems. This produces lower susceptibility to social prediction errors but reduces the capacity for the cooperative inference that high social precision affords.

Sensory Anchoring Prior

The sensory anchoring prior governs the relative precision weighting assigned to the foundational sensory channels the agent uses as its ultimate evidential ground truth. This is the informational basis to which the generative model appeals when resolving higher-tier prediction errors, ultimately defining the informational thickness of the Markov blanket. *Exteroceptive anchoring* reflects a baseline trust directed outward. The system prioritises environmental states like visual, auditory, and spatial cues to validate its internal models. The self is constituted in relation to external spatial coordinates and exteroceptive feedback, such that environmental regularities serve as the primary scaffolding for self-model stability.

Interoceptive anchoring reflects a baseline trust directed inward. The continuous monitoring of cardiac, respiratory, and visceral signals forms the bedrock of reality-testing for the self-model. The minimal self is tightly bound to internal homeostatic and physiological rhythms, and selfhood is primarily grounded in the body's internal milieu.

This dimension bears directly on individual differences in interoceptive sensitivity and their documented relationship to affective and somatic psychopathology (Craig, 2002; Seth, 2013).

Epistemic Meta-Priors

Epistemic meta-priors govern how the agent prefers to minimise free energy within the constraints established by Tiers 1 and 2. Where the upper tiers set the parameters of the agent's fundamental orientations, Tier 3 concerns the characteristic strategies deployed in navigating prediction errors across the perception-action cycle. They are "meta-priors" because they constitute beliefs about the optimal policy for updating beliefs, acting as second-order priors governing how the generative model itself should be managed.

Precision Bias

Precision bias concerns the relative weighting assigned to internal beliefs versus incoming prediction errors. A *prior-heavy, top-down bias* reflects high precision on internal beliefs relative to sensory evidence. The agent minimises free energy by assimilating incoming signals into existing models, filtering out contradictory data, or acting to bring the world into alignment with its predictions. This configuration favours model stability and coherence at the cost of sensitivity to genuine environmental change.

A *sensory-heavy, bottom-up bias* reflects high precision on incoming prediction errors relative to prior beliefs. The agent minimises free energy by rapidly revising internal models to accommodate new evidence. This configuration favours environmental tracking at the cost of stability, and may render the agent vulnerable to destabilising belief updates in conditions of high sensory noise.

Action-Perception Precision

Active and perceptual inference occur simultaneously and cannot be explicitly selected between; individuals nonetheless differ systemati-

cally in the relative precision assigned to each when confronted with free energy. An *action-heavy* configuration places high precision on the agent's capacity to alter external states. The primary mode of free energy minimisation is active environmental transformation, such as exerting control, manipulating circumstances, or forcing external reality into conformity with the internal model. This maps onto external locus of control and behavioural dominance, and intersects with the differentiation drive at Tier 2.

A *perception-heavy* configuration places lower precision on action efficacy relative to sensory updating. The agent minimises free energy primarily through belief revision, attentional reorientation, and accommodation of the environment rather than action upon it. This maps onto internal locus of control and accommodative cognitive styles, and in its extreme form may contribute to behavioural inhibition when the tractability prior is insufficient to sustain action-oriented strategies.

Epistemic Foraging Drive

The epistemic foraging drive governs the expected information gain the agent assigns to sampling high-variance, boundary-testing states. An agent with a *high foraging drive* actively seeks environments, whether somatic or semantic, that inject prediction errors into the system. It expects that such errors, once processed, will reduce long-term uncertainty and prevent overfitting of the generative model. This is the active inference analogue of sensation-seeking (Zuckerman, 1994). An agent with a *low foraging drive* prefers highly predictable, low-variance environments and minimises voluntary exposure to uncertainty.

The CPC Model distinguishes two functionally separable subtypes of foraging, hypothesised to be partially orthogonal rather than a single continuum. *Somatic foraging* involves testing the physical Markov blanket through biological edge-seeking. This means injecting prediction errors into the physiological system to prove the existence and resilience of the minimal, bodily self. At its

mild end, somatic foraging characterises voluntary perturbations of bodily sensing through exercise, breathwork, or interoceptive practice. At its extreme, it encompasses visceral threat-seeking, such as extreme sports or edge-play, using the unambiguous certainty of intense somatic sensation to ground and confirm the self-model.

Semantic foraging involves testing the conceptual and cultural architecture of the agent's generative model by probing the coherence of its worldview under pressure. Mild semantic participant foraging is characteristic of engagement with transgressive art, radical philosophy, or complex esoteric and symbolic systems. Extreme semantic foraging may involve deliberate phenomenological destabilisation through psychedelics or other profound altered states of consciousness. This functions as a directed attempt to observe the underlying structure of the generative model by shattering the conceptual ego and collapsing its surface representations.

This mechanism is formalised by the REBUS (Relaxed Beliefs Under Psychedelics) model (Carhart-Harris & Friston, 2019), which proposes that classical serotonergic psychedelics reduce the precision weighting of high-level priors. This process effectively flattens the free energy landscape and shallows the attractor basins that ordinarily maintain stable prior configurations. Within the CPC framework, the therapeutic and transformative potential attributed to psychedelic experience may reside precisely in this landscape-flattening. By temporarily reducing attractor depth at the level of Tier 1 and Tier 2 priors, these agents open pathways for prior reorganisation that would otherwise remain energetically inaccessible. Importantly, the depth of the agent's characteristic attractor, which is a direct function of prior precision accumulated across development, is expected to modulate the response. An agent with a very deep attractor basin may require greater perturbation to achieve meaningful reorganisation than one whose prior landscape is comparatively shallow.

Temporal Depth

Temporal depth concerns the functional time horizon over which the agent's active inference operates. An agent with *shallow temporal depth* optimises primarily for immediate sensory feedback and homeostatic regulation. This is a reactive mode in which the generative model is evaluated against near-term outcomes. An agent with *deep temporal depth* simulates cascading future states and optimises for allostasis, entering a proactive mode in which action selection is governed by anticipated long-term model accuracy rather than immediate error reduction. Temporal depth is structurally constrained by the terminal prior at Tier 1: an agent with active cessation-seeking or passive disengagement has, by definition, a truncated effective temporal horizon, rendering deep temporal inference structurally unavailable as a strategy.

Self-Model Coherence

Self-model coherence concerns the temporal stability and narrative integration of the agent's highest-level self-referential priors, focusing on the precision and temporal consistency of the beliefs the agent holds about itself across successive inference cycles. This dimension maps onto the philosophical distinction between the *minimal self*, the immediate, pre-reflective agent constituted at the boundary of its Markov blanket, and the *narrative self*, the temporally extended, autobiographically integrated identity maintained across time (Gallagher, 2000).

Self-model coherence, in the CPC formulation, concerns principally the narrative pole. It evaluates the degree to which the agent's self-referential priors are consolidated into a stable, temporally coherent structure capable of absorbing local perturbations without wholesale revision. *Low coherence* reflects a self-model that is revised rapidly and discontinuously in response to local prediction errors. The agent's identity shifts to accommodate each new context, social input, or affective state without integration into a stable narrative structure.

tic configurations across the lower-tier dimensions. Five latent factors are hypothesised, each representing a theoretically coherent syndrome of co-varying dimensions expected to load together on factor-analytic assessment. These factors are summarised in Table 1.

These five factors carry clear functional correspondences to existing descriptive models. Epistemic Plasticity maps broadly onto Openness to Experience; Cybernetic Stability maps onto the inverse of Neuroticism, with affective valence as its phenomenological felt correlate; Blanket Permeability maps onto the affiliative pole of Agreeableness and related communion constructs; and Temporal Extension maps onto Conscientiousness. Self-Model Stability constitutes a theoretically novel factor not cleanly captured by any single Big Five dimension, intersecting with both Neuroticism and Conscientiousness while carrying distinct clinical implications. The CPC Model's claim is not that these five factors replace the Big Five but that they identify the generative prior architecture from which Big Five phenotypes emerge as behavioural expressions.

Relationship to Existing Models

The CPC Model is positioned as a generative complement to the primary existing frameworks in personality science, rather than a competitor to them. Its relationship to each is considered in turn.

Big Five Personality Theory

The Big Five traits (Goldberg, 1990; McCrae & John, 1992) represent the most extensively validated descriptive taxonomy of personality. The CPC Model proposes to explain the mechanisms generating the variance those dimensions capture. As noted above, Epistemic Plasticity corresponds broadly to Openness to Experience; Cybernetic Stability to the inverse of Neuroticism, with affective valence as the phenomenological felt correlate of tractability-prior configuration; Blanket Permeability to Agreeableness-related constructs; and Tem-

poral Extension to Conscientiousness. Extraversion, however, does not map cleanly onto any single CPC dimension. It likely emerges from an interaction of the tractability prior, which drives approach motivation, and the dissolution drive, which motivates social engagement and affiliation. This non-correspondence is itself a prediction of the model: if Extraversion reflects a combination of prior parameters rather than a single prior dimension, it should not map cleanly onto any one CPC factor, and its distinctive cross-factor loading pattern follows accordingly.

Hierarchical Taxonomy of Psychopathology (HiTOP)

The Hierarchical Taxonomy of Psychopathology (HiTOP, Kotov et al., 2017) has advanced a dimensional, structural model of mental illness that supersedes traditional categorical diagnoses. The CPC Model shares HiTOP's hierarchical architectural ethos and provides a generative mechanism for its empirically derived spectra. Within the CPC framework, the broad spectra of HiTOP, such as Internalizing and Externalizing, can be conceptualised as the phenotypic output of specific hyperprior configurations. For instance, the Internalizing spectrum aligns with the downstream consequences of a low tractability prior combined with high expected volatility (producing chronic expected free energy), whereas the Externalizing spectrum may reflect the combined outputs of a high differentiation drive and a prior-heavy precision bias. By mapping HiTOP's descriptive hierarchies onto the CPC's generative prior hierarchy, psychopathology can be understood not merely as statistical deviations, but as rigid, self-reinforcing cybernetic attractors.

BIS/BAS Theory

Gray's (1982) Behavioural Inhibition and Activation Systems provide a neurobiological account of approach and avoidance motivation. The tractability prior directly formalises the BIS/BAS distinction within a generative modelling framework.

Figure 1

Generative architecture of Trait Neuroticism as an illustrative Core Prior Cascade. Cybernetic states are represented as parameter distributions positioned on dimensional axes across the prior hierarchy. The depicted profile illustrates a characteristic Neuroticism-associated configuration: high expected volatility, low tractability, and low uncertainty tolerance cascading through elevated expected free energy and a perception-heavy action-perception ratio to produce negative affective valence and high trait Neuroticism at the observable phenotypic level. Colours indicate functional tiers.

Generative Architecture of Trait Neuroticism

Cybernetic states represented as parameter distributions across the prior hierarchy

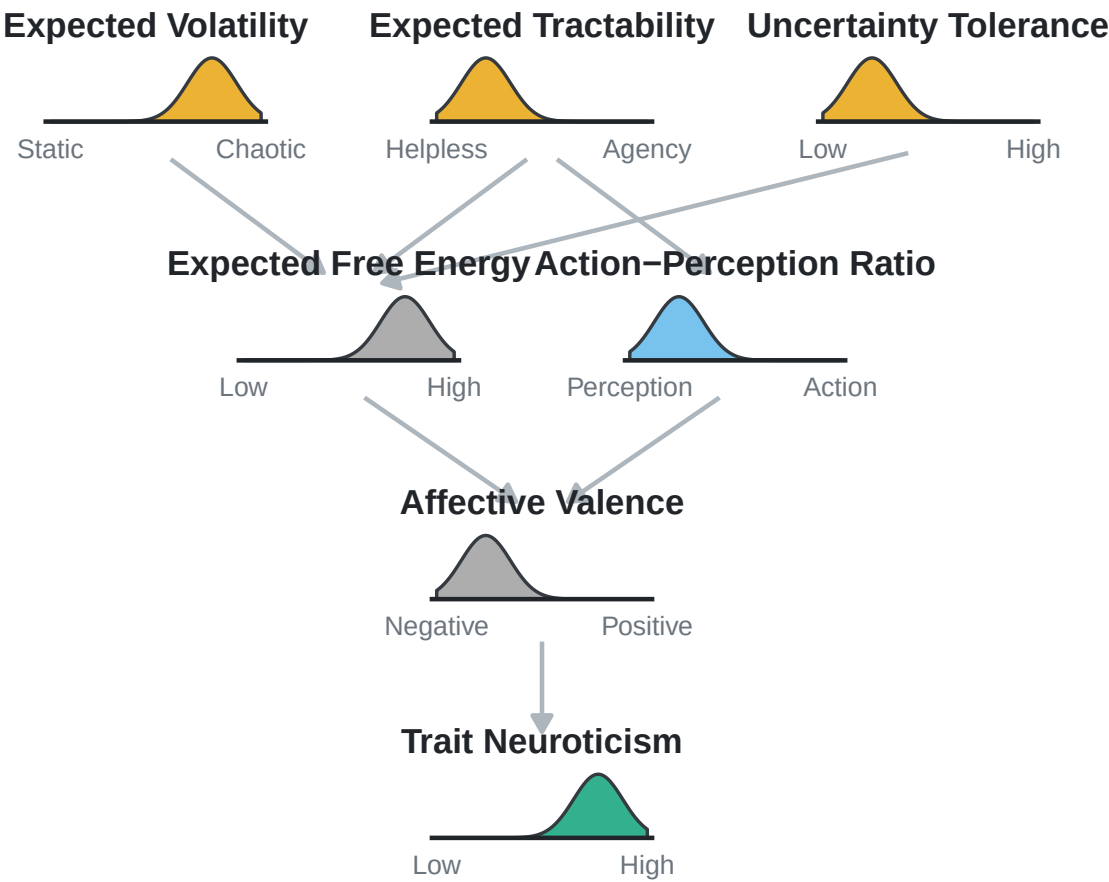


Table 1

Hypothesised five-factor structure of the Core Prior Cascade Model. Each factor represents a theoretically coherent syndrome of co-varying prior dimensions.

Factor	Primary Construct	Correlated Dimensions
Factor 1: Epistemic Plasticity	How the agent handles new information	Sensory-heavy precision bias; high epistemic foraging drive; high uncertainty tolerance
Factor 2: Cybernetic Stability	Baseline expectation of safety versus threat	High tractability prior; positive affective valence; low volatility prior (negative correlate: epistemic shielding)
Factor 3: Blanket Permeability	How the self relates to the group	High dissolution drive; high social inference precision
Factor 4: Temporal Extension	The timescale of the agent's generative model	Deep temporal depth; terminal transcendence; high differentiation drive
Factor 5: Self-Model Stability	Temporal integration and coherence of the self-model	High self-model coherence; high tractability prior; deep temporal depth (negative correlates: low uncertainty tolerance; active cessation-seeking)

High tractability maps onto dominant BAS functioning, characterised by sustained approach, reward sensitivity, and behavioural activation. Low tractability maps onto dominant BIS functioning, characterised by behavioural inhibition, hypervigilance to potential prediction errors, and an anxiety-prone response to uncertainty. The added theoretical value is mechanistic. BIS/BAS describes the behavioural output; the CPC Model proposes the hierarchical prior structure that produces it, grounding the distinction in the mathematics of expected free energy minimisation and extending it with the upstream Tier 1 parameters (volatility prior, uncertainty tolerance) that modulate BIS/BAS expression.

Attachment Theory

The permeability prior and its associated Tier 2 dimensions map directly onto the attachment

styles identified by Bowlby (1969) and Ainsworth. Avoidant attachment corresponds to a high differentiation drive with low social inference precision. The agent minimises expected free energy by maintaining rigid blanket boundaries and discounting social prediction errors. Anxious-preoccupied attachment corresponds to a high dissolution drive combined with high social inference precision and low tractability. The agent urgently seeks high coupling with attachment figures but cannot trust the tractability of that coupling to resolve relational uncertainty. Secure attachment corresponds to a calibrated permeability prior and high tractability. The agent can engage in flexible coupling because it believes the resulting prediction errors will be resolvable. The CPC formulation offers a mechanistic account of why insecure attachment styles persist: they are self-consistent configurations of interacting priors that are self-reinforcing across develop-

ment, because the prior structure shapes the environments sampled and the interpretations assigned to those environments.

Sensation Seeking

Zuckerman's (1994) sensation-seeking construct overlaps directly with the epistemic foraging drive. The novel contribution of the CPC framework is the proposed distinction between somatic and semantic foraging as two potentially orthogonal subtypes. Existing sensation-seeking scales conflate physical risk-taking (somatic foraging) with intellectual novelty-seeking (semantic foraging) within a single dimension. The CPC Model hypothesises these to be partially separable, predicting that the two subtypes will show discriminant validity. This hypothesis constitutes a direct, testable empirical contribution requiring dedicated scale development and factor-analytic verification.

Cybernetic Big Five Theory

DeYoung's (2015) Cybernetic Big Five Theory (CB5T) represents the most direct theoretical antecedent of the CPC Model. CB5T conceptualises the Big Five in cybernetic terms, organising traits around two meta-traits, Plasticity (exploration and flexibility) and Stability (goal-maintenance and threat response), grounded in generic reward and threat parameters. The CPC Model builds on this framework by operationalising the underlying parameters within the formal mathematics of the FEP rather than the more informal vocabulary of cybernetics. Where CB5T describes an organism that "reacts to threat," the CPC Model specifies that the organism experiences a chronically elevated expected free energy signal (negative affective valence) arising from a low tractability prior within a high-volatility environment. DeYoung's Plasticity meta-trait corresponds to the behavioural output of CPC Factor 1, and his Stability meta-trait to CPC Factor 2. The CPC Model thus offers CB5T a formal mathematical grounding, extending it with dimensions that lie outside CB5T's current scope:

the terminal prior, the sensory anchoring prior, and the somatic/semantic foraging distinction.

Jungian Archetypes

Jungian archetypes can be formalised within the CPC framework as deeply entrenched, phylogenetically conserved hyperpriors, serving as low-dimensional structural attractors in the human generative model that are inherited rather than individually acquired (Jung, 1968). As high-precision priors embedded at the apex of the hierarchy, archetypes function as strange attractors. They normally operate in the background, shaping the solution space for higher-level inference without themselves becoming objects of explicit awareness. During episodes of extreme semantic foraging, when the agent deliberately injects prediction errors into its conceptual architecture, the surface-level generative model destabilises. As superficial models collapse, the system falls back on these deeper phylogenetic attractors to prevent catastrophic free energy increase. An archetypal encounter, on this account, constitutes the agent's experience of reaching the bedrock of the human generative model: the ancient, conserved structural constraints that represent the species' collective prior.

REBUS Model and Psychedelic Research

The REBUS (Relaxed Beliefs Under Psychedelics) model (Carhart-Harris & Friston, 2019) provides a pharmacologically grounded account of how the hierarchical prior structure proposed by the CPC Model can be temporarily altered through neurobiological intervention. REBUS proposes that classical serotonergic psychedelics act via the 5-HT_{2A} receptor to reduce the gain on descending prediction signals from high-level cortical areas, effectively flattening the free energy landscape at the level of its deepest priors. Within the CPC architecture, this corresponds to a reduction in the precision of Tier 1 existential hyperpriors and Tier 2 boundary priors, which under normal conditions form the

deepest and most change-resistant attractor basins. As the landscape temporarily flattens, prediction errors that would ordinarily be suppressed by these high-precision priors are permitted to propagate upward through the hierarchy. This destabilises surface-level models and, under favourable set and setting conditions, enables reorganisation around new prior configurations. The therapeutic benefits reported in psychedelic-assisted treatment of depression, addiction, and post-traumatic stress disorder may thus be understood as the functional consequence of transient attractor-basin shallowing. The agent is temporarily liberated from a deeply entrenched, pathologically stable prior configuration and afforded the opportunity to settle into an adjacent basin with lower expected free energy.

The CPC Model generates specific predictions for REBUS-relevant research. Individual differences in prior depth, operationalised as the precision weighting of Tier 1 hyperpriors, are expected to moderate psychedelic response. Agents with deeper attractor basins (e.g., those with high Cybernetic Stability or highly entrenched low-tractability priors) may require greater or more sustained perturbation to achieve comparable reorganisation. Conversely, agents with shallower or more volatile prior landscapes (e.g., high Epistemic Plasticity, low self-model coherence) may be more susceptible to destabilisation without achieving coherent re-organisation, consistent with observations of adverse psychedelic responses in vulnerable individuals.

Limitations and Open Questions

The Core Prior Cascade Model offers several contributions to the theoretical landscape of personality science. First, by grounding individual differences in the mathematics of the Free Energy Principle, it provides a principled mechanistic account of why personality configurations are stable, why they co-vary in the patterns observed by descriptive models, and why certain configurations confer differential psychopathological vulnerability. The

cascade architecture implies that interventions targeting Tier 3 parameters (as in most cognitive and behavioural therapies) will have limited durability unless Tier 1 and Tier 2 priors are also addressed, since deep stability and chronic suffering are maintained by the prior configuration at the highest levels of the hierarchy, not by surface-level epistemic strategies.

Second, the distinction between somatic and semantic epistemic foraging constitutes a theoretically motivated prediction not generated by existing models. If confirmed by discriminant validity evidence, this distinction would have implications for understanding heterogeneity within sensation-seeking, risk tolerance, and creative openness, and for the differential phenomenology of body-anchored versus concept-anchored approaches to self-exploration.

Third, the terminal prior provides a formal active inference account of temporal self-extension and its pathological extremes. The active cessation-seeking pole of the disengagement configuration, in particular, offers a mechanistic reframing of suicidal ideation as a specific prior configuration. Rather than treating it as merely a symptom category, it conceptualizes ideation as the agent's rational minimisation of expected free energy given a prior structure that renders all available future states unresolvably costly.

Despite these contributions, several limitations and open questions remain. The most pressing empirical challenge is developing valid measurement instruments capable of assessing the proposed prior parameters. Traditional subjective questionnaires predominantly capture the “observables” of personality. They do not directly access the unconscious priors that form the actual scaffolding of the human conscious experience. Because these deeply entrenched generative models operate outside introspective awareness, self-reports merely capture their downstream phenomenological output.

However, this measurement gap is not insurmountable. By reliably measuring these observables and rigorously analysing their covariance, re-

searchers can make robust inferences about the higher-order structural causes generating them. The latent variables modeled from dense, high-quality psychometric data can serve as proxies for the deeper prior landscape.

Furthermore, implicit behavioural tasks offer a promising alternative to self-report. Because explicit questionnaires are vulnerable to introspective limits, tasks that probe inference processes directly may be better suited for assessing CPC dimensions. As demonstrated by Makowski (?), who found systematic links between personality traits and illusion sensitivity in a perceptual decision task, implicit measures can bypass declarative awareness to probe the underlying predictive architecture directly. A battery of perceptual, somatic, and cognitive tasks could eventually serve as direct assays of the CPC Model's latent dimensions. Computational psychiatry provides relevant methodological precedents here (Huys et al., 2016), where parameters analogous to the CPC's Tier 3 dimensions (learning rates, reward sensitivity, temporal discounting) have been successfully back-estimated from task performance using hierarchical Bayesian and reinforcement-learning models.

The CPC framework extends this tradition by situating these parameters within an explicit prior hierarchy, generating the prediction that different computational task designs will probe different tiers. Paradigms involving estimation of environmental volatility will tax Tier 1 parameters; tasks involving social inference and boundary detection will probe Tier 2 priors; and tasks requiring rapid strategy-switching under conflicting evidence will engage Tier 3 epistemic meta-priors. Paradoxically, the most informative tier, Tier 1 hyperpriors, will be the hardest to access directly. Their influence operates precisely by suppressing the prediction errors that would otherwise reveal them. Paradigms that systematically manipulate volatility, tractability, and uncertainty across extended task sequences may be best positioned to back-out these deep prior parameters.

Additionally, the proposed five-factor structure

is hypothetical and awaits empirical validation. Factor-analytic assessment will be necessary to determine the actual latent structure of co-variation across CPC dimensions, and it is plausible that the factor structure will vary across populations and developmental periods. Future work should prioritise the development of a compact self-report or behavioural instrument capable of indexing the key CPC dimensions, the design of implicit and computational modelling paradigms for prior parameter estimation, and longitudinal investigation of how prior structure evolves across development and in response to therapeutic intervention.

References

- Bakan, D. (1966). *The duality of human existence: An essay on psychology and religion*. Rand McNally.
- Barrett, L. F. (2017). The theory of constructed emotion: An active inference account of interoception and categorization. *Social Cognitive and Affective Neuroscience*, 12(1), 1–23.
- Becker, E. (1973). *The denial of death*. Simon; Schuster.
- Bowlby, J. (1969). *Attachment and loss: Vol. 1. attachment*. Basic Books.
- Carhart-Harris, R., & Friston, K. J. (2019). REBUS and the anarchic brain: Toward a unified model of the brain action of psychedelics. *Pharmacological Reviews*, 71(3), 316–344.
- Craig, A. D. (2002). How do you feel? Interoception: The sense of the physiological condition of the body. *Nature Reviews Neuroscience*, 3(8), 655–666.
- DeYoung, C. G. (2015). Cybernetic big five theory. *Journal of Research in Personality*, 56, 33–58.
- Erikson, E. H. (1968). *Identity: Youth and crisis*. W. W. Norton & Company.
- Eysenck, H. J. (1967). *The biological basis of personality*. Charles C Thomas Publisher.
- Fleeson, W., & Jayawickreme, E. (2015). Whole trait theory. *Journal of Research in Personality*, 56, 82–92.
- Friston, K. (2010). The free-energy principle: A

- unified brain theory? *Nature Reviews Neuroscience*, 11(2), 127–138.
- Friston, K. (2019). A free energy principle for a particular physics. *arXiv Preprint arXiv:1909.10863*.
- Gallagher, S. (2000). Philosophical conceptions of the self: Implications for cognitive science. *Trends in Cognitive Sciences*, 4(1), 14–21.
- Goldberg, L. R. (1990). An alternative "description of personality": The big-five factor structure. *Journal of Personality and Social Psychology*, 59(6), 1216–1229.
- Gray, J. A. (1982). *The neuropsychology of anxiety: An enquiry into the functions of the septo-hippocampal system*. Clarendon Press.
- Greenberg, J., Pyszczynski, T., & Solomon, S. (1986). The causes and consequences of a need for self-esteem: A terror management theory. *Public Knowledge, Private Ignorance: Toward a Resolution of Cognitive Dissonance*, 189–212.
- Huys, Q. J. M., Maia, T. V., & Frank, M. J. (2016). Computational psychiatry as a bridge from neuroscience to clinical applications. *Nature Neuroscience*, 19(3), 404–413.
- Jung, C. G. (1968). *The archetypes and the collective unconscious*. Princeton University Press.
- Kotov, R., Krueger, R. F., Watson, D., Achenbach, T. M., Althoff, R. R., Bagby, R. M., Brown, T. A., Carpenter, W. T., Caspi, A., Clark, L. A., et al. (2017). The hierarchical taxonomy of psychopathology (HiTOP): A dimensional alternative to traditional nosologies. *Journal of Abnormal Psychology*, 126(4), 454.
- Marcia, J. E. (1966). Development and validation of ego-identity status. *Journal of Personality and Social Psychology*, 3(5), 551–558.
- McAdams, D. P. (1996). Personality, modernity, and the storied self: A contemporary framework for studying persons. *Psychological Inquiry*, 7(4), 295–321.
- McCrae, R. R., & John, O. P. (1992). An introduction to the five-factor model and its applications. *Journal of Personality*, 60(2), 175–215.
- Mischel, W., & Shoda, Y. (1995). A cognitive-affective system theory of personality: Reconceptualizing situations, dispositions, dynamics, and invariance in personality structure. *Psychological Review*, 102(2), 246.
- Seligman, M. E. P. (1972). Learned helplessness. *Annual Review of Medicine*, 23(1), 407–412.
- Seth, A. K. (2013). Interoceptive inference, emotion, and the embodied self. *Trends in Cognitive Sciences*, 17(11), 565–573.
- Zuckerman, M. (1994). *Behavioral expressions and biosocial bases of sensation seeking*. Cambridge University Press.